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11PLUSDIYTABMA-ST-001

11PlusDIY: MA-ST-001

Mathematics Mock Exam

Exam Name: MA-ST-001

INSTRUCTIONS TO CANDIDATES

- Write your name, exam date, exam name and exam centre in the spaces provided on the answer booklet provided. Please write clearly and in capital letters.
- **Do not write on this exam, use the answer booklet provided.**
- Use black ink
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Place an X in your choice of answer. Only choose one.
- If you make a mistake. Draw a line clearly through the incorrect answer.

A	
B	X
C	
D	
E	

Correct

A	
B	X
C	
~~D~~	~~X~~
E	

Correct

A	X
B	X
C	√
D	X
E	X

Incorrect

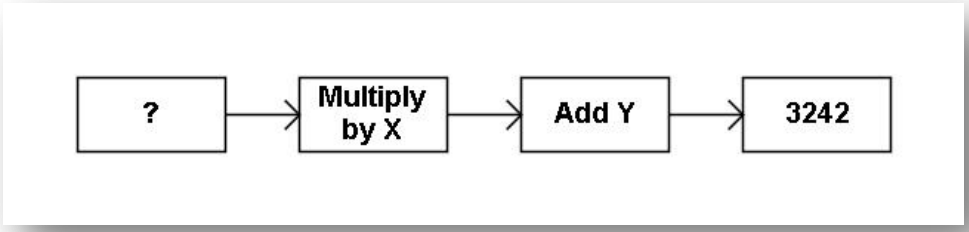
A	X
B	
C	
D	X
E	

Correct if two answers are required

A	X
B	
C	
D	
E	

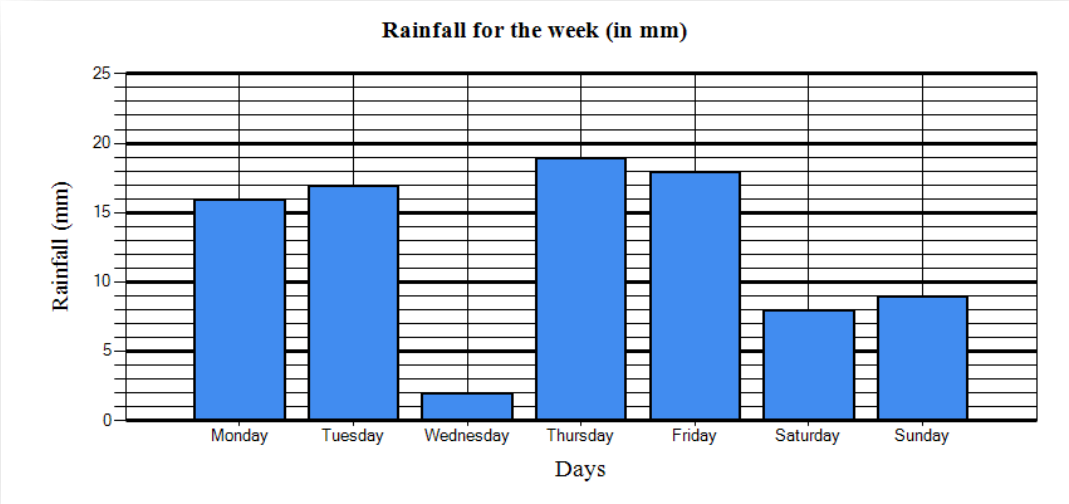
Incorrect

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1	Look at the fraction below and fill in the missing number. $\frac{8}{10} - \frac{6}{16} = \frac{[?]}{40}$	13 26 12 17
2	What is the Lowest Common Multiple of 9, 3 and 4	25 13 36 22
3	Add Forty Seven Thousand, Four Hundred and Twenty Eight with Twenty Two Thousand, Six Hundred and Thirty Six	70100 70064 70015 70085
4	Multiply Thirty One with Seven Hundred	21700 21731 21745 21733
5	Subtract Nine Thousand, Five Hundred and Sixty Six from Fifty Two Thousand, Five Hundred and Nine	42934 42909 42914 42943
6	In the number machine shown below if X = 3 and Y = 407 then what number was put into the machine to start with? 	936 954 943 945
7	Subtract 568.72 and 455.499	113.977 113.715 113.221 113.421

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8	Divide 79.82 by 32.5	2.456 3.3160 3.256 3.4548
9	Which of the following decimals is the SMALLEST? [61.4, 34.3, 47.80, 54.8, 31.186]	34.3 54.8 47.80 61.4 31.186
10	The chart below shows the rainfall recorded during the week. What was the total rainfall recorded during the week? 	89 94 83 82
11	Represent 53/625 as a percentage	7.38% 6.90% 8.90% 8.48%
12	Reduce the fraction : 924/972	76/79 77/77 77/81 72/78

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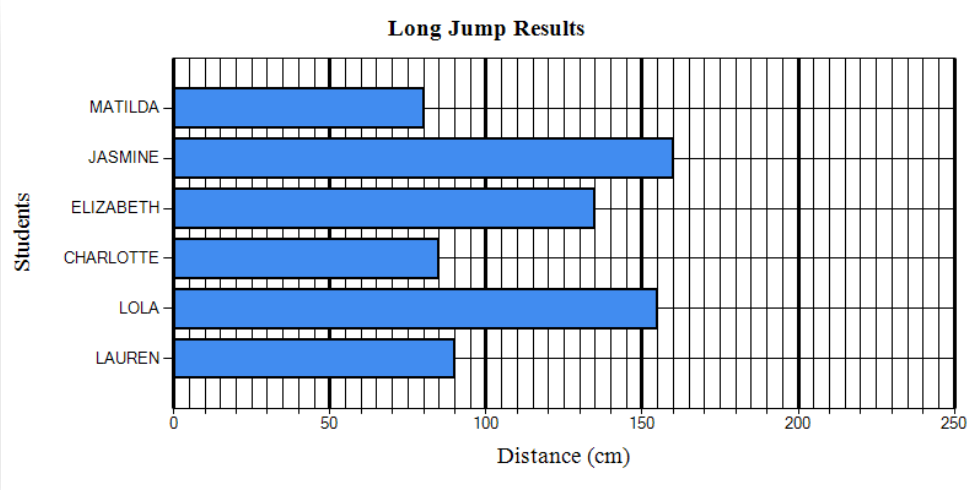
13	A Builder used SQUARE tiles to cover a rectangular surface. Each side of the square tile was 50 cm and he used 612 tiles to cover the surface. There were 34 tiles along one of the sides of the surface. What was the PERIMETER of the surface covered by the tiles?	52 cm 52 m 36.44 m 14.9 cm
14	LEXI is baking a Pound cake. The recipe requires her to mix FLOUR and BUTTER in the ratio 3:8. If LEXI uses 201 gms of FLOUR then how much of BUTTER should she put in the Pound cake?	608 gms 487 gms 536 gms 421 gms
15	The table below shows the prices for chocolates at a shop. SNICKERS 37 pence JELLY BABIES 52 pence SMARTIES 61 pence M&MS 53 pence CURLY WURLY 33 pence MEGAN has £7.00 to do the shopping. If MEGAN buys 5 M&MS and 3 CURLY WURLY then how much money is MEGAN left with?	£4.37 £3.36 £4.13 £0.98
16	Last Friday 4% of the employees in a factory were off sick and 16% were away on training. 250 employees work in the factory. How many employees were available for work?	212 198 204 200
17	While on a holiday to Croatia LACEY purchased a toy for 21 Croatian Kuna. The exchange rate is £1 = 8.57 Croatian Kuna. What is the cost of the toy in UK pounds?	£2.45 £4.10 £2.80 £0.18
18	COURTNEY, SCARLETT and TILLY are each given fifteen SOFT MINTS from a bag and there are four left over. How many SOFT MINTS were originally in the bag?	49 51 46 50

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19	KATIE, ZARA, MARTHA and SCARLETT are all members of the school 4 x 400 meters relay team. Their individual timings (in minutes and seconds) in a 400 meters run are 01:05, 01:50, 01:45 and 01:40 in that order. Because of the practice they are doing the team is expected to better their collective relay race time by 11 seconds. What is the best time the team can hope for in the competition?	04:10 ☐ 06:09 ☐ 05:15 ☐ 05:50 ☐
20	The table below shows the temperatures recorded in cities around the world. **Senegal - Dakar** 17 deg C **Angola - Luanda** 8 deg C **Nigeria - Abuja** 5 deg C **Zambia - Lusaka** 34 deg C **Canada - Ottawa** 0 deg C **Qatar - Doha** 26 deg C **Greece - Athens** 4 deg C **Italy - Rome** 15 deg C Which city was the hottest?	Nigeria - Abuja ☐ Angola - Luanda ☐ Zambia - Lusaka ☐ Greece - Athens ☐
21	Four cars take part in a long distance road race. The total distance to be covered is 420 Kilometres. The average speeds of the cars are shown in the table below in Kilometres per hour. **Car A** 120 kmph **Car B** 180 kmph **Car C** 60 kmph **Car D** 168 kmph What distance has the slowest car in the race covered when the fastest car crosses the finish line?	140 km ☐ 153 km ☐ 139 km ☐ 130 km ☐
22	The names of students in a class are listed below. If a student was selected from the list at random then what is the probability that their name **STARTS-WITH 'C'**? (FAITH, AMBER, CHLOE, LILY, COURTNEY, CHARLOTTE, ALEXANDRA, CAITLIN, SUMMER, SKYE)	20/17 ☐ 2/5 ☐ 10/7 ☐ 20/19 ☐
23	The tally chart below shows the chocolate preferred by students in a school. Each symbol represents 15 students ($ = 15 students). **DOUBLE DECKER** $ **MINT AERO** $ $ $ $ $ $ **TWIX** $ $ $ $ **GALAXY CARAMEL** $ $ $ $ $ $ $ $ $ **AERO** $ $ **MINSTRELS** $ $ $ $ $ $ $ $ How many students like AERO?	120 ☐ 15 ☐ 150 ☐ 30 ☐

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24 The graph below shows the results of long jump competition at a local school in centi-meters (cm).



How much further did the winner of the competition jump as compared to student who came last?

80 cm	
69 cm	
92 cm	
93 cm	

25 Look at the number series below.

27, 18, 11, 44, [?], 12, 5, 32, 21, 28, 39

If the median of the series is 25, then what is the missing number?

30	
25	
33	
29	

26 The village fete has a choloclate luck dip where you are allowed to pick one of the chocolates in the box. The table below shows the mix of chocolates in the lucky dip box.

BUTTONS	19
CARAMAC	20
MOW WOAMS	28
MARS BAR	10
MILKA	9

What is the probability of picking MOW WOAMS from the dip?

2/25	
13/50	
14/43	
22/133	

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27 The table below shows the distances between different towns in Km (kilo meters).

-	HERMITAGE	TILEHURST	DISTINGTON	DUMBARTON	WATCHETTS	THURSO
HERMITAGE	0	84	69	50	79	96
TILEHURST	84	0	98	66	77	69
DISTINGTON	69	98	0	24	60	50
DUMBARTON	50	66	24	0	82	30
WATCHETTS	79	77	60	82	0	63
THURSO	96	69	50	30	63	0

GRACIE travelled from HERMITAGE to TILEHURST and then from TILEHURST to WATCHETTS. How many KM did GRACIE cover in total?

- 141 ☐
- 177 ☐
- 161 ☐
- 163 ☐

28 NICOLE, GRACIE and LAUREN all bought some sweet. NICOLE has 21 sweets. GRACIE and LAUREN together have a total of 51 sweets. NICOLE and GRACIE together have 49 sweets. How many sweets do the children have in total?

- 73 ☐
- 74 ☐
- 71 ☐
- 72 ☐

29 The sum of THREE consecutive numbers is 117. What is the smallest of these three numbers?

- 40 ☐
- 37 ☐
- 39 ☐
- 38 ☐

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30	ISLA is making a STRAWBERRY SHORTCAKE from a recipe which is in imperial units. The recipe requires 17 Oz of Flour. Assuming 1 Oz = 30 gms and rounding to the nearest 5 gms, how much Flour should ISLA use?	510	
		105	
		195	
		525	

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Important 2011 Mock Exam Dates	
Birmingham	21 August 2011 (Sold Out)
	11 September 2011 (Sold Out)
	18 September 2011 (Sold Out)
	23 October 2011
London	11 September 2011
	25 September 2011
	24 October 2011 (HALF TERM)
West Yorkshire	25 September 2011

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Answers

1	Answer: 17 Explanation: "$\frac{8}{10} - \frac{6}{16} = \frac{17}{40}$ "
2	Answer: 36 Explanation: "Factors of 9 are 1, 3, 9. Factors of 3 are 1. Factors of 4 are 1, 2, 4."
3	Answer: 70064 Explanation: "47428 + 22636 = 70064"
4	Answer: 21700 Explanation: "31 × 700 = 21700"
5	Answer: 42943 Explanation: "52509 - 9566 = 42943"
6	Answer: 945 Explanation: "(945 x 3) + 407 = 3242"
7	Answer: 113.221 Explanation: "568.72 - 455.499 = 113.221"
8	Answer: 2.456 Explanation: "79.82 ÷ 32.5 = 2.456"
9	Answer: 31.186 Explanation: "The decimals in ascending order are - 31.186, 34.3, 47.80, 54.8, 61.4"
10	Answer: 89 Explanation: "16 + 17 + 2 + 19 + 18 + 8 + 9 = 89"
11	Answer: 8.48% Explanation: "53/625 = 8.48%"
12	Answer: 77/81 Explanation: "The common factor is 12. 924/972 = 77/81"
13	Answer: 52 m Explanation: "Length of first side is = Tile Side x No of tiles used = 50 x 34 The length of second side can be calculated by dividing the total tiles used by the number of tiles along one side. The two sides therefore are 1700 and 900 cm"
14	Answer: 536 gms Explanation: "Amount of BUTTER = (201 / 3) * 8 gms"

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15	Answer: £3.36 Explanation: "Cost for 5 M&MS = £2.65. Cost for 3 CURLY WURLY = £0.99. Total cost = £3.64"
16	Answer: 200 Explanation: "4% of 250 is 10. 16% of 250 is 40. Therefore total employees available for work = 250 - (10 + 40) = 200."
17	Answer: £2.45 Explanation: "Cost in Pounds = 21 / 8.57 = £2.45"
18	Answer: 49 Explanation: "There are Three children and each gets Fifteen sweets. The total sweets therefore = (3 x 15) + 4 = 49"
19	Answer: 06:09 Explanation: "(01:05 + 01:50 + 01:45 + 01:40) - 11 seconds = 06:09"
20	Answer: Zambia - Lusaka Explanation: "34 was the maximum temperature"
21	Answer: 140 km Explanation: "The speed of the fastest car is 180 kmph. Time taken by fastest car to complete the race = 420 / 180 = 2 hrs 20 min. The speed of the slowest car is 60 kmph. Distance covered by the slowest car in the same time = 140 km"
22	Answer: 2/5 Explanation: "The names of students in a class are listed below. If a student was selected from the list at random then what is the probability that their name **STARTS-WITH'C'**? (FAITH, AMBER, CHLOE, LILY, COURTNEY, CHARLOTTE, ALEXANDRA, CAITLIN, SUMMER, SKYE). The probability is therefore 2/5"
23	Answer: 30 Explanation: "Symbols x Multiple = 2 x 15 = 30"
24	Answer: 80 cm Explanation: "Max = 160 cm. Min = 80 cm. The range = 160 - 80 = 80 cm."
25	Answer: 25 Explanation: "The median is given. This median has to be one of the numbers in the series."
26	Answer: 14/43 Explanation: "28 / Sum of all chocolates = 28/86 = 14/43"
27	Answer: 161 Explanation: "Distance from HERMITAGE to TILEHURST is 84 km and from TILEHURST to WATCHETTS is 77 km. Total = 161 km"
28	Answer: 72 Explanation: "NICOLE has 21 sweets, GRACIE has 28 sweets and LAUREN has 23 sweets."

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29	Answer: 38 Explanation: "If N is the middle number of the series then, N + (N -1) + (N + 1) = 3N = 117. Thus N -1 = 38"
30	Answer: 510 Explanation: "1 oz = 30 gms, therefore 17 Oz = 17 x 30 = 510 gms"

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